Azure Blob Storage – Access Tier and Lifecycle Management

Objective

To create an Azure Blob Container, upload a sample file, change its access tier, and configure Lifecycle Management to optimize storage costs.

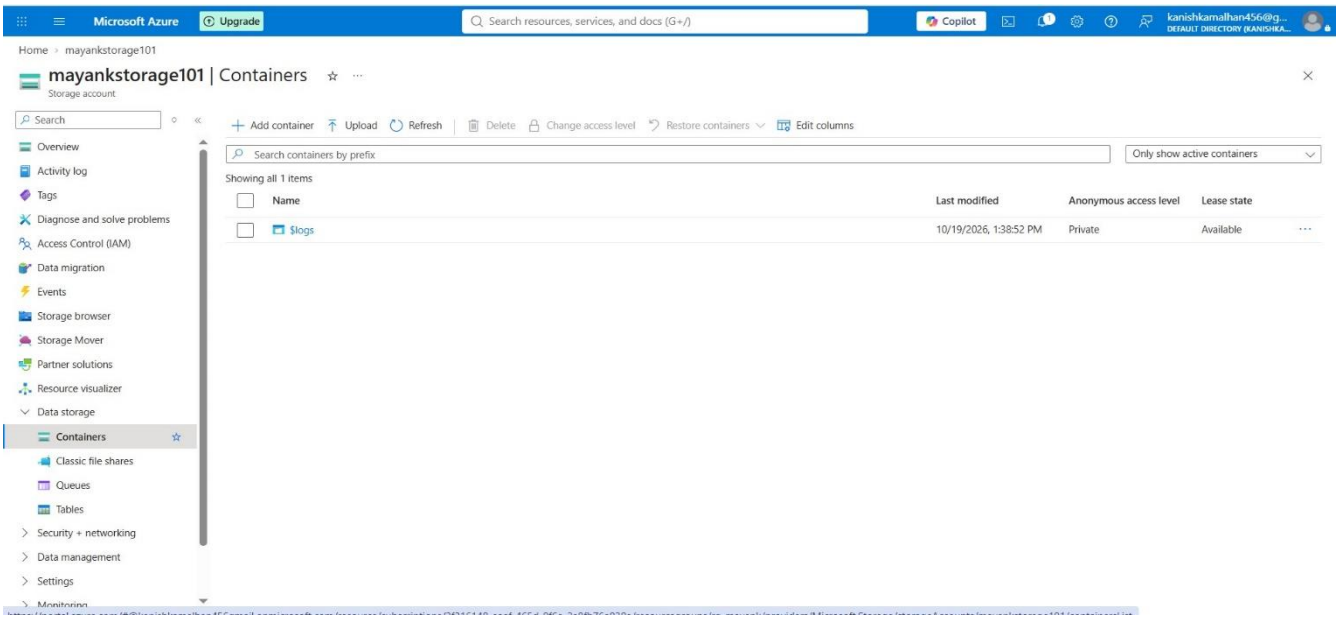
Prerequisites

- An active Azure subscription.
- An Azure Storage Account named `mayankstorage101`.
- A sample text file to upload.

Procedure

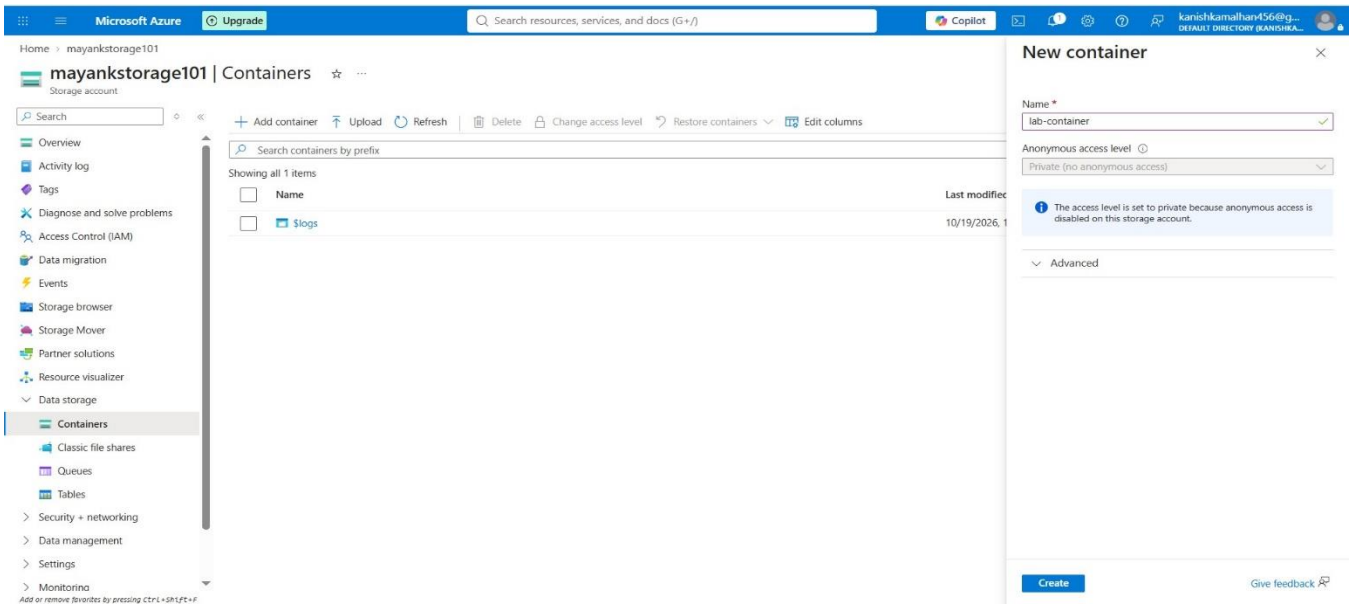
Step 1: Open the Storage Account Containers

Open the Azure Portal and navigate to the `mayankstorage101` Storage Account. Select Containers under Data storage to view the existing containers.



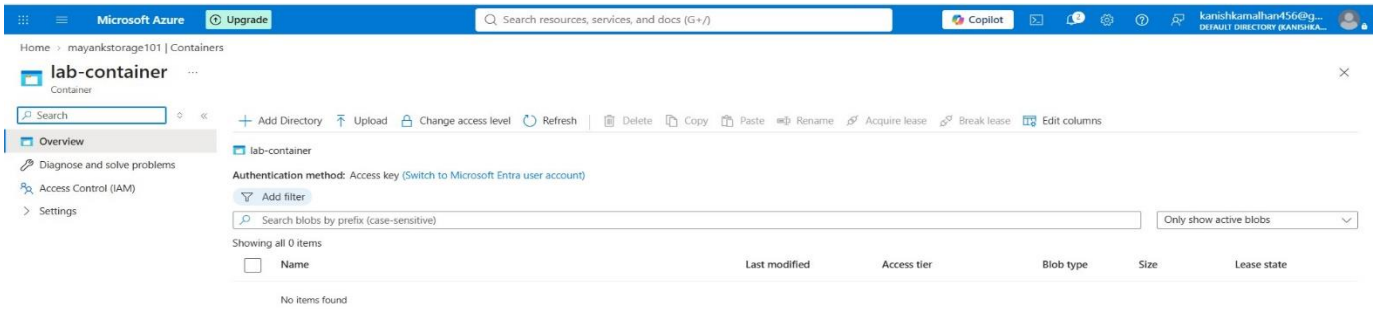
Step 2: Create a New Container

Click Add container. Enter `lab-container` as the container name and keep the anonymous access level set to Private. Click Create.



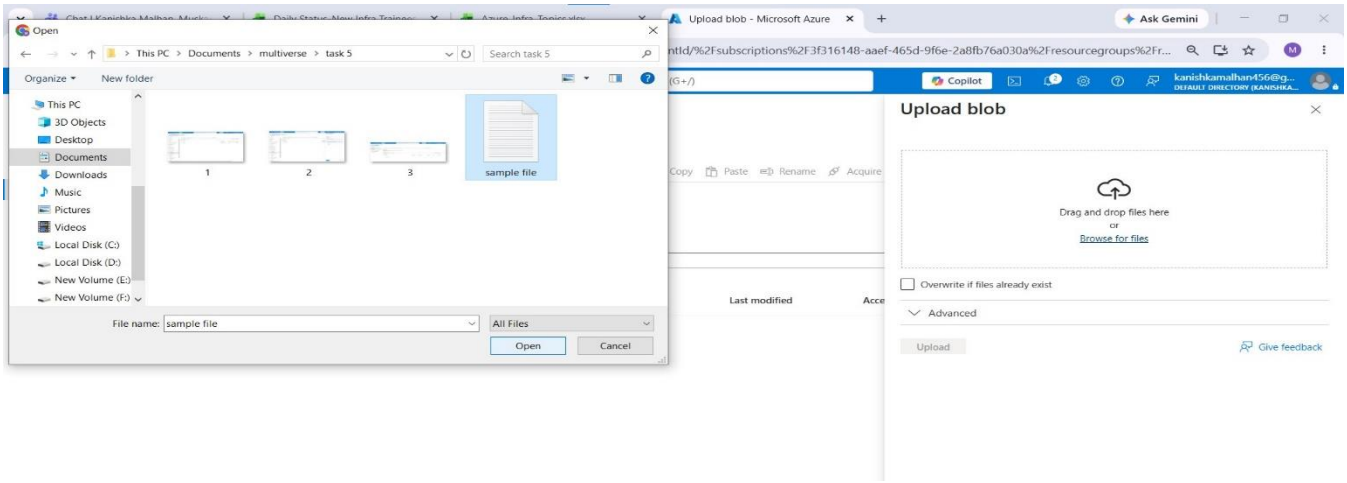
Step 3: Verify the Container

After creating the container, open `lab-container`. Verify that the container is available and ready to store blob objects.



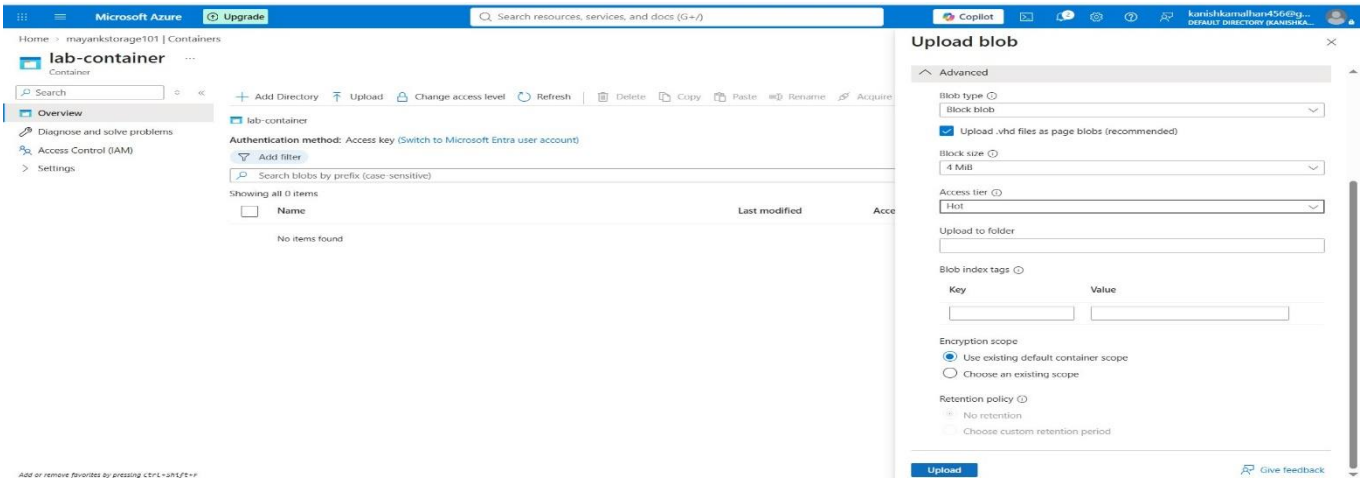
Step 4: Select a File for Upload

Click Upload inside the container. In the file selection window, browse your computer and select the sample text file.



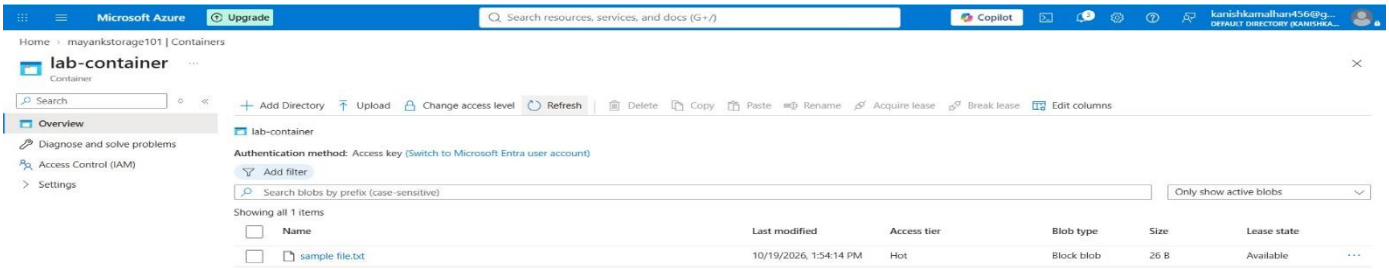
Step 5: Configure the Blob Upload

In the Upload blob panel, verify the selected file and review the upload settings. Keep the container private and configure the access tier as required for the lab. Click Upload.



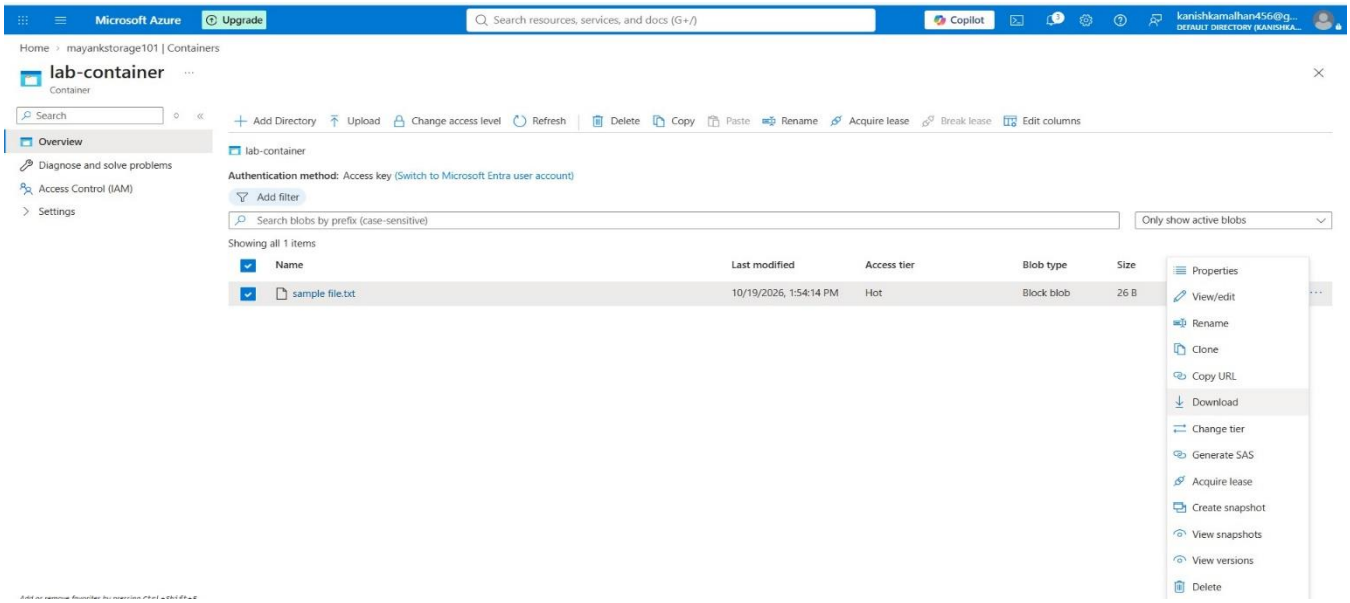
Step 6: Verify the Uploaded Blob

After the upload completes, check the container contents. Confirm that the uploaded file or files appear in the blob list.



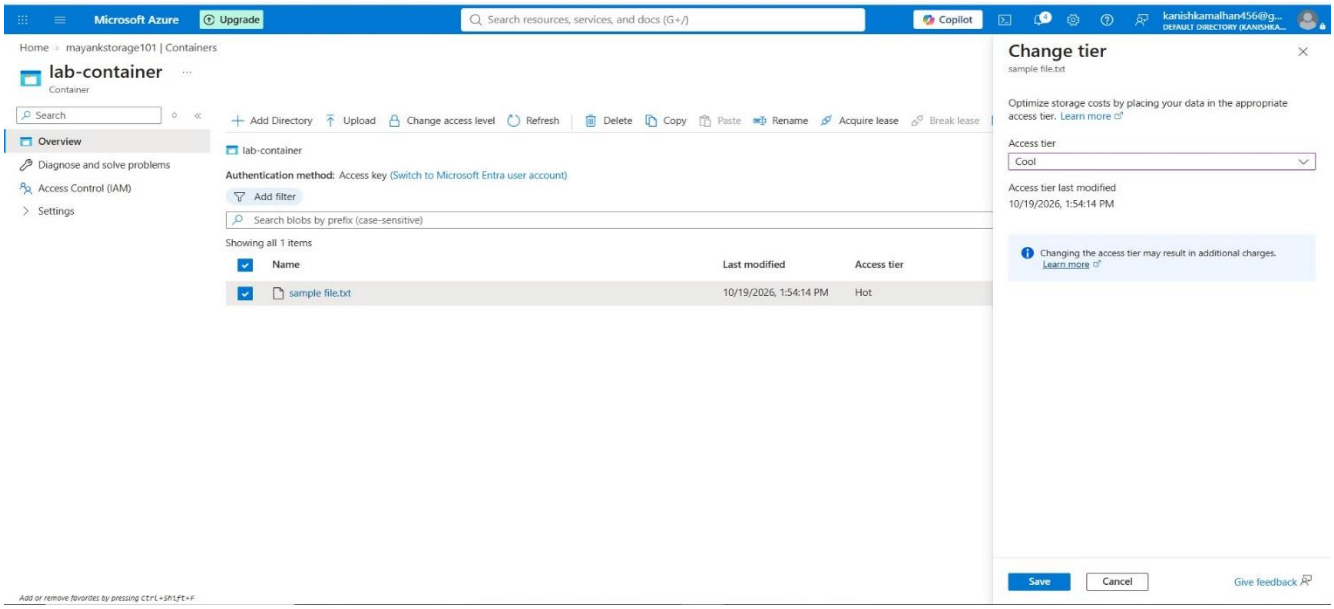
Step 7: Open the Blob Actions

Select the required blob and open its More options menu. Choose the option to change its access tier.



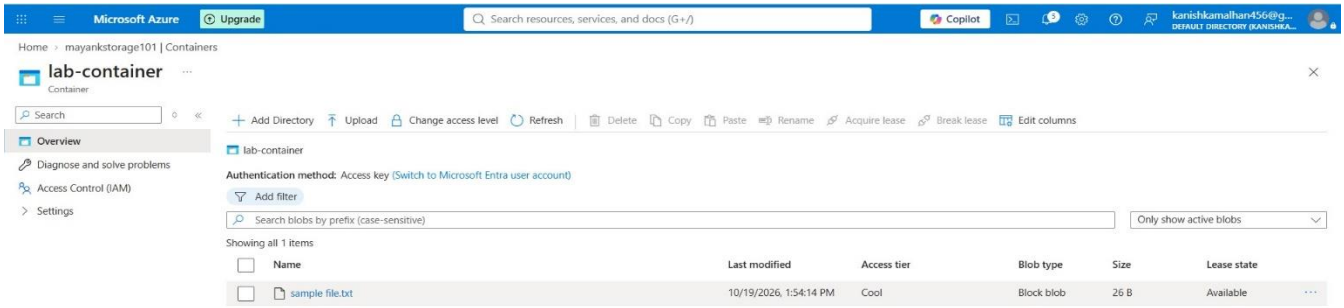
Step 8: Change the Blob Access Tier

In the Change tier panel, select the required storage tier for the blob and confirm the operation. Azure Blob Storage provides Hot, Cool, Cold, and Archive tiers, subject to account and blob support.



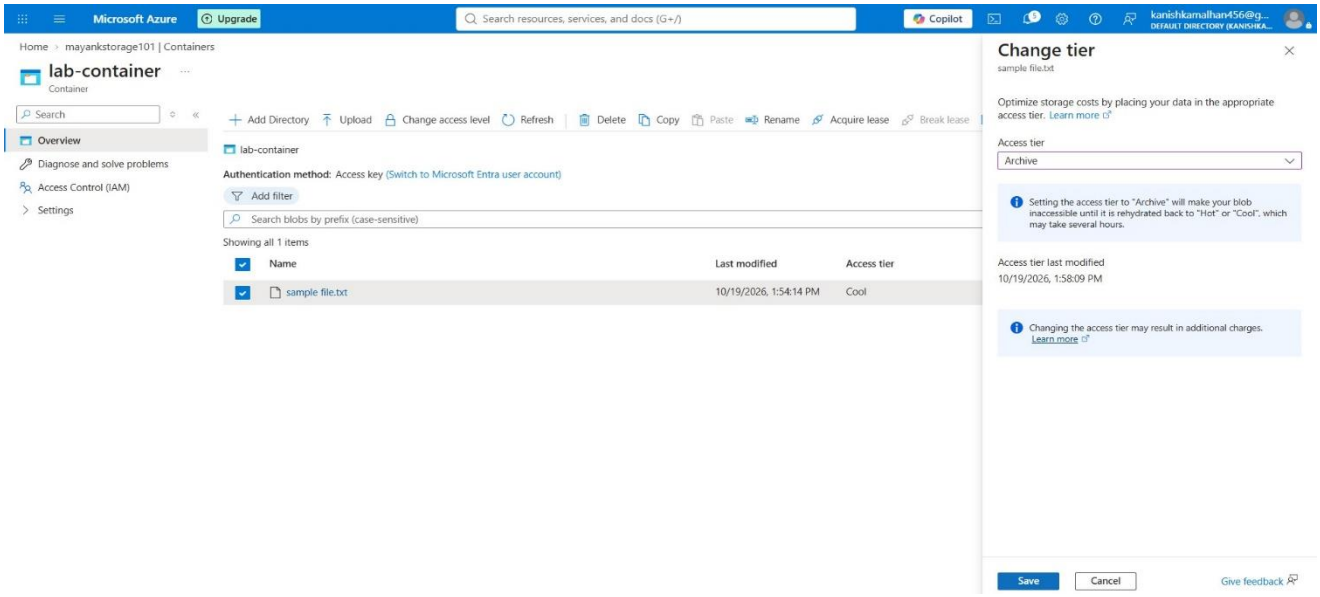
Step 9: Verify the Blob Tier to cool

Return to the container and inspect the blob list. Check the file's displayed access tier to verify the result of the tier-change operation.



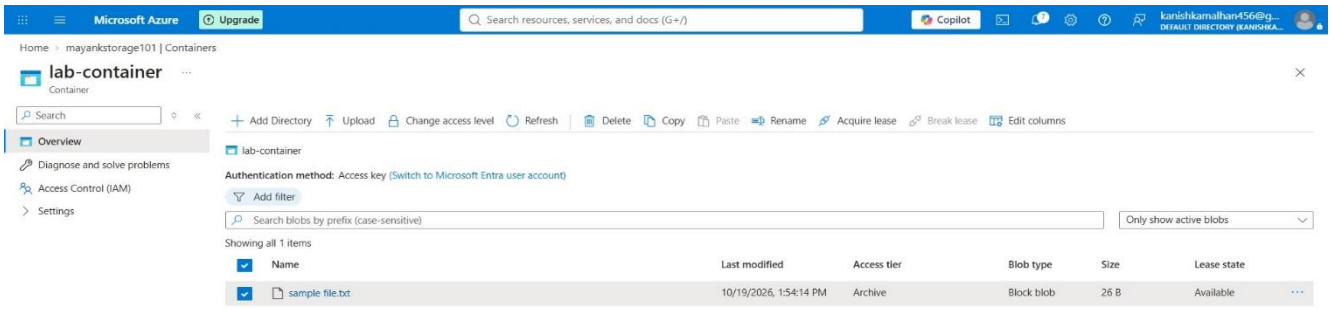
Step 10: Perform Another Tier Change to Archive

Open the Change tier panel for the selected blob and choose the target tier shown in your screenshot. Confirm the operation to apply the change.



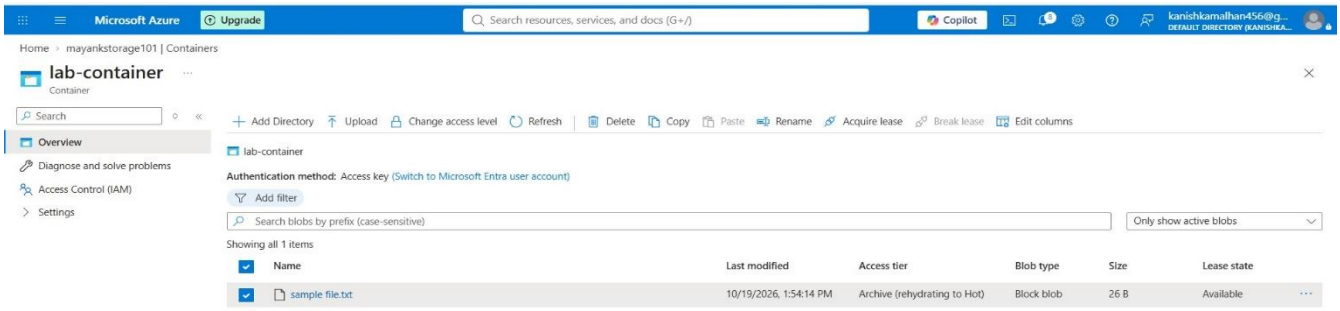
Step 11: Review the Blob Access Tier

Open `lab-container` and review the blob details, including the file name, last modified date, access tier, blob type, and size.



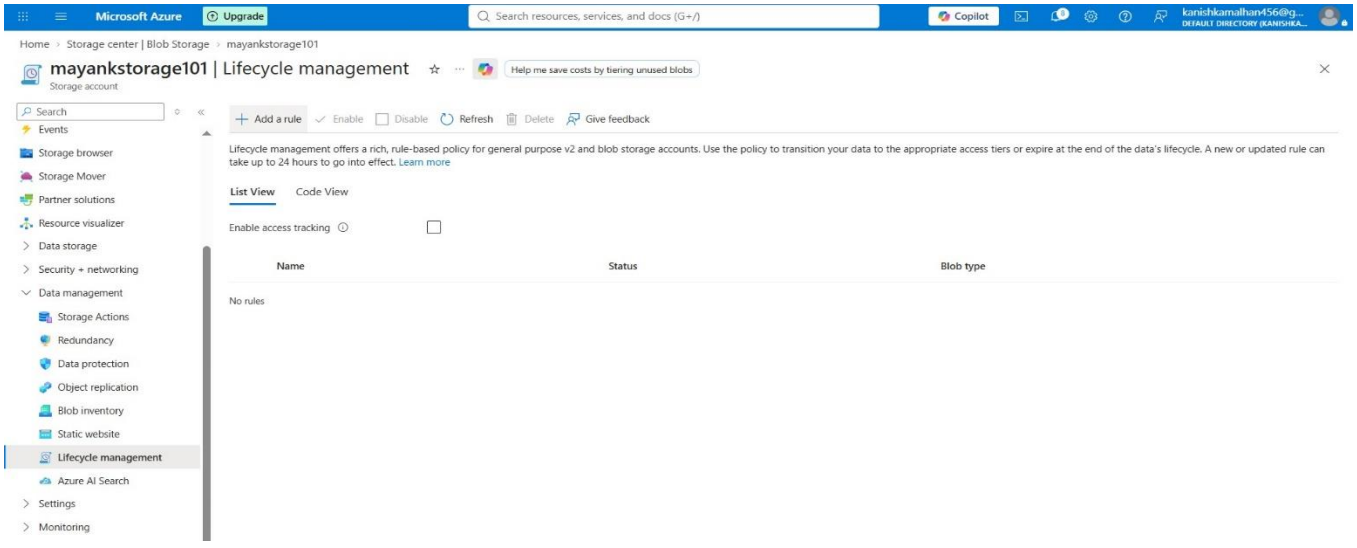
Step 12: Verify the Current Blob Status

Refresh the container view and verify the displayed access tier and lease state of the sample blob. If the blob is in the Archive tier, it cannot be read directly until it is rehydrated.



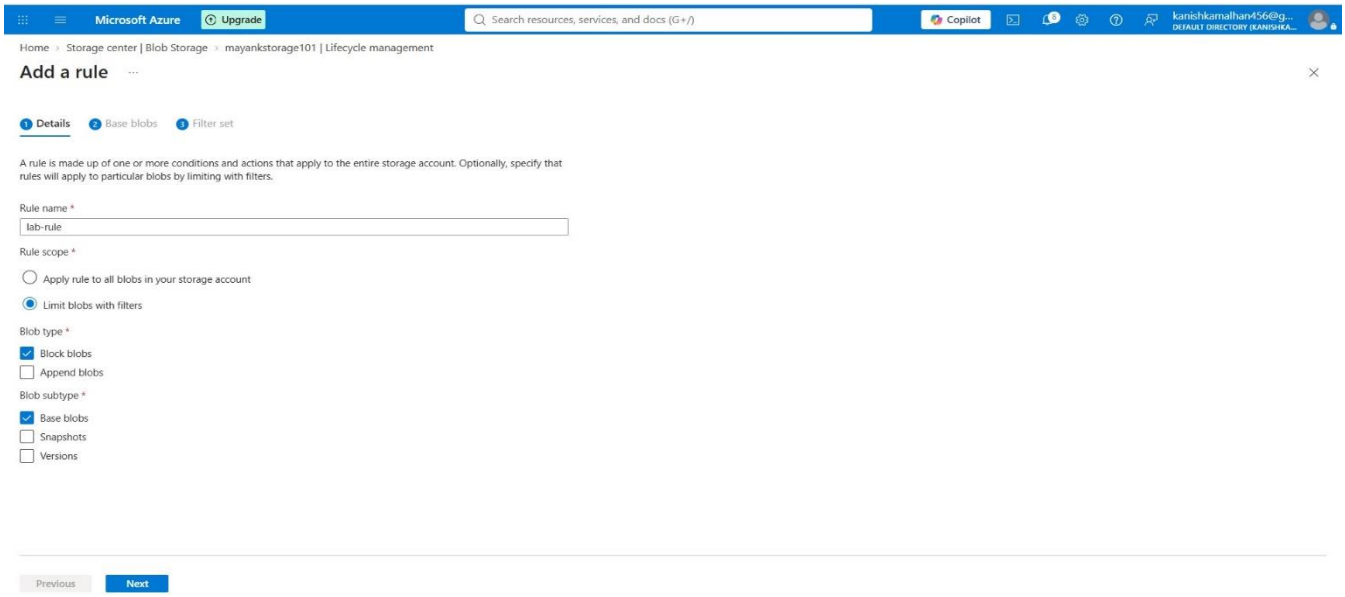
Step 13: Open Lifecycle Management

Navigate to the `mayankstorage101` Storage Account. Under Data management, select Lifecycle management and click Add a rule.



Step 14: Configure the Lifecycle Rule Details

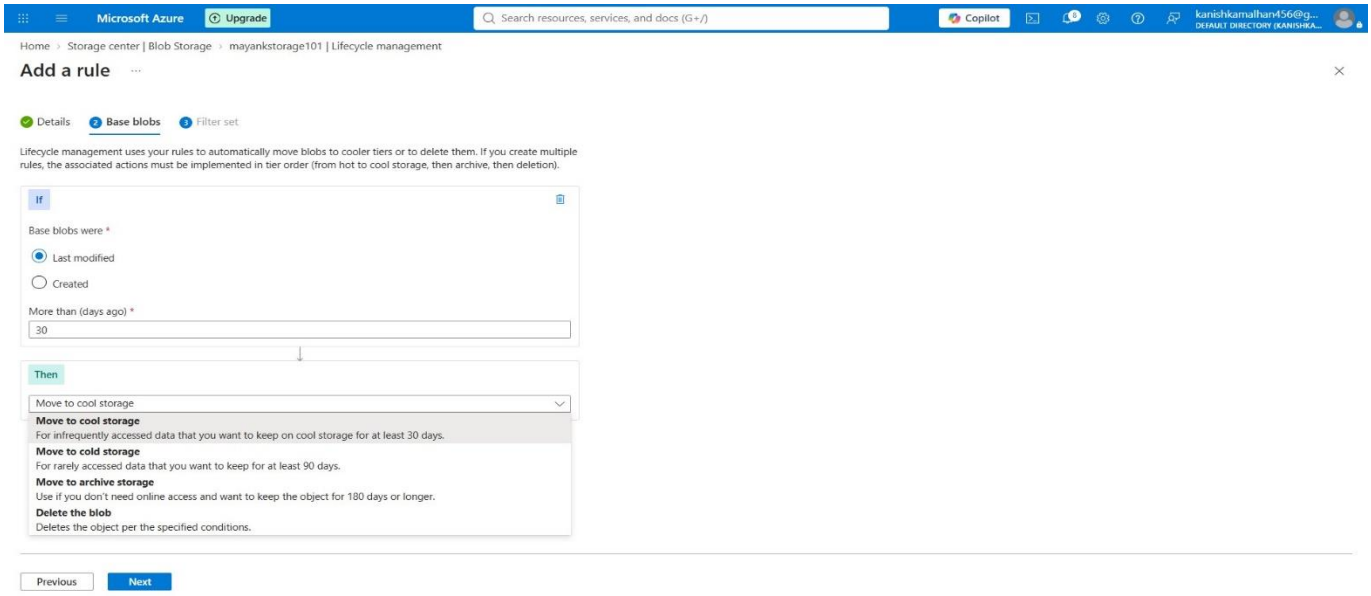
Enter `lab-rule` as the rule name. Select Limit blobs with filters, choose Block blobs, and select Base blobs. Click Next.



Step 15: Configure the Lifecycle Condition and Action

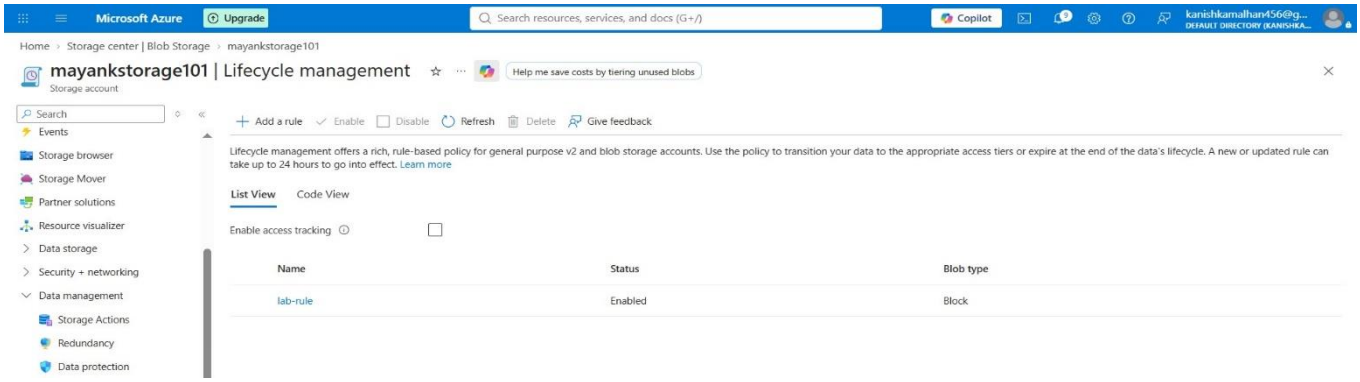
Under Base blobs, select Last modified and enter `30` days. Under the Then section, select Move to cool storage. Click Next to continue.

This rule is intended to move eligible blobs to the Cool tier when their last modification was more than 30 days ago.



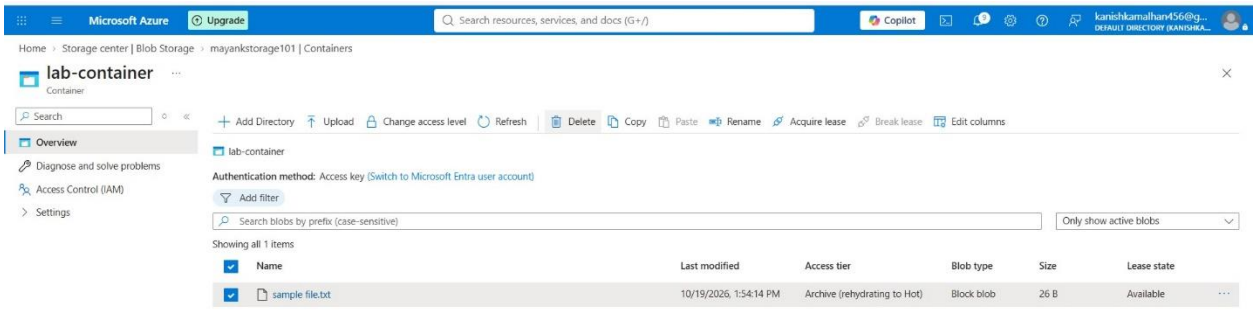
Step 16: Verify the Lifecycle Rule

After completing the remaining configuration and creating the rule, return to the Lifecycle management page. Verify that `lab-rule` appears with the status Enabled.



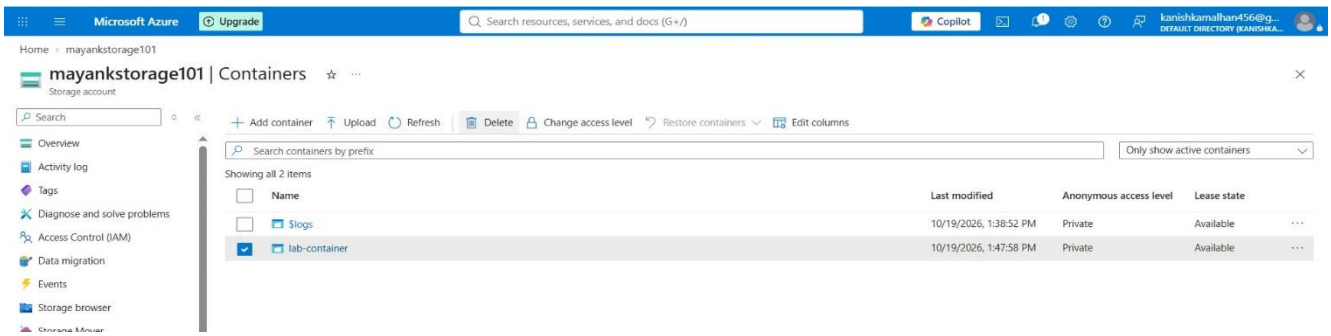
Step 17: Delete the Blob File After the Operation

Delete the file by click on delete option.



Step 18: Delete the Container in the Storage Account

Delete the blob container , select the container and click on delete



Result

A blob container was created, sample files were uploaded, and blob access tier operations were performed. A Lifecycle Management rule named `lab-rule` was also created and enabled with a condition based on the last modified date.